

INSTITUTE OF ACTUARIES OF INDIA

EXAMINATIONS

SP8 - General Insurance: Pricing
May 2025

INDICATIVE SOLUTION

Introduction

The indicative solution has been written by the Examiners with the aim of helping candidates. The solutions given are only indicative. It is realized that there could be other points as valid answers and examiner have given credit for any alternative approach or interpretation which they consider to be reasonable.

Q. 1) a) Principles of Insurance-

- Insurable interest eg: You can not purchase insurance for neighbour's car
- Quantifiable & Financial loss eg: Loss of your favourite toy is an emotional loss and hence cant be insured
- Limited financial loss eg: restore the insured to their pre-loss financial position, not to profit from the loss
- Low probability event of loss eg: certain definitive events aren't covered under insurance
- Homogeneity or Pooling of risks eg: underwrite a large no of similar risks as the probability of claims from all the insured at the same time is quite remote
- Independent risks eg: Multiple heart attacks are not covered in Critical Illness policy as post first one, other tend to follow
- Low Moral Hazard eg: Physical damage of mobile phone is not insurable as it may be self-induced after newer models are launched
- Any other valid point e.g subrogation, contribution, loss minimisation etc.

(Max 4)

b) Reimbursement of maternity expenses is not a valid form of insurance, unless it results into some emergency complications as it is a planned event

(2)

[6 Marks]**Q. 2)****a)**

Policy number	Sum Insured (a)	Premium amount (b)	Claim amount (c)	Insurer's Loss Ratio = (c) / (b)	Cession % (d) = (a) / Limit	RI Premium (e) = (d) * (b)	RI Claims (f) = (d) * (c)	RI Viability = (f) - (e)
Policy 1	750,000	15,000	13,000	87%	33%	5,000	4,333	(667)
Policy 2	1,000,000	20,000	25,000	125%	50%	10,000	12,500	2,500

Policy 1 is profit making for insurer and RI viability is also negative, hence policy 1 is not viable. Similarly, policy 2 is viable under the RI arrangement.

(7)

b) Retrocession is an arrangement of reinsurance purchased by Reinsurer to further cede its risk underwritten

(1)

[8 Marks]**Q. 3) a) Perils**

- Fire in forest
- Illness of animals
- Animal attack to visitors or other animals
- Insufficient visitors to ensure profitability
- Any other valid Peril

(2)

b) insurance products

- Fire or Property insurance
- Medical insurance for animals or Pet insurance
- Public liability insurance

- Business interruption insurance
- Any other valid insurance product

(2)

c) Risk Factors

- Number of animals
- Type of animals
- Type of enclosure
- No. of visitors each day
- Liability in case of visitor attack
- Health of animals
- Age of animals
- Availability and quality of Health care provider service available
- Any other valid risk factor

(3)

d) Rating factors are characteristics or proxies of insured risks which are objective, measurable and verifiable. Risk factors are attributes of risk which may or may not qualify as rating factors.

(1)

e) Rating factors

- Number of animals
- Type of animals
- Type of enclosure
- No. of visitors each day
- Ticket cost or VIP tickets proportion
- Age of animals
- Past medical history of animals
- No. of doctors on site
- Any other valid rating factor

(4)

[12 Marks]

Q. 4)

a) Captive insurance is an insurer that is wholly owned by an industrial or commercial enterprise and set-up with the primary purpose of insuring the parent or associated group companies and retaining premiums and risk within the enterprise.

The term captive insurance company is often loosely applied to an insurance company set-up to sell insurance to the parent's customers. Thus, a Bank or NBFC having their own insurance company mainly to insure its customers e.g. vehicle insurance for vehicle financed.

(3)

b) Main reasons for captives-

- Focusing effort on risk management
- Managing the overall insurance spend
- Providing direct access to the reinsurance market
- Providing insurance cover not available else
- Any other valid reason

(3)

[6 Marks]**Q. 5) Areas of regulatory interventions for General Insurance in India-**

- Maintaining minimum Solvency requirement
- Restricting investments in less risky assets
- Maintaining minimum rating for Reinsurers and Credit houses
- Pricing based on objective and consistent criteria
- Offering all necessary coverages as prescribed by regulator
- Charging minimum rates for tarified products
- Matching tenure, value and currency of assets to liabilities
- Sharing periodic prescribed disclosures
- Paying levies to required consumer bodies
- Using licensed agents to sell products post adequate training
- Ensuring minimum prescribed level of customer service
- Restriction on individuals holding key positions in insurance companies
- Any other valid point

[10 Marks]**Q. 6) Economic factors affecting low-speed E-bikes:**

- Inflation may be higher than expected
 - Labour fees depend on wage inflation
 - Cost of replacement parts depends upon price inflation
 - Indemnity of liability towards third party depends upon court inflation and/or medical inflation
- Higher commissions may be paid due to inflation or levels of competition
- High level of taxes or duties may have reduced primary sales of e-bikes, thus reducing demand of this insurance product
- Low level of underwriting cycle leading to low rates and low profitability conditions
- Legislative changes towards selling this product may have changed
- Any other valid point

[Max 5 Marks]**Q. 7) Shortcomings of traditional burn cost approach-**

- Model error or parameter error
- Random fluctuations / lack of stochastic approach
- External sources like demand pattern changes or economic factor changes
- Changes in competitor strategy
- Regulatory changes
- Change in claims management procedure
- Different distribution channel
- Different mix of business
- Varied reinsurance rates
- More than expected expenses
- Any other valid point

[Max 5 marks]**Q. 8)**

- a) Actuarial value addition in Reinsurance department
 - Quantifying risk appetite of the Company

- Analysing exposure information
- Estimating areas of accumulation & associated risks
- Structure required reinsurance cover
- Negotiate on Reinsurer rates based on ceding portfolio
- Provide reserving inputs at a net level
- Calculate residual risk for capital modelling
- Any other valid point

(Max 4)

b)

Claim No.	Gross Loss	Net of Prop Treaty 85% of Gross Loss	Net of Risk XoL min (Net of Prop Treaty, 20L)	Net of Aggregate XoL Agg. Upto 25 L
Claim 1	5,000,000	4,250,000	2,000,000	2,000,000
Claim 2	1,000,000	850,000	850,000	2,500,000
Claim 3	350,000	297,500	297,500	2,500,000
Claim 4	150,000	127,500	127,500	2,500,000

(Max 6)

c) Reasons why Facultative cover is better:

- Only Claim1 seems to be an outlier as a large claim
- So, purchasing a treaty to cover outlier risks may be expensive
- Facultative cover will allow reducing deductible and cover for all other risks, so more of smaller risks can be ceded
- If a facultative cover is purchased, lesser amount will inure of the Aggregate XoL treaty. So, it will be used for effectively
- It creates more capacity with insurer to write more business
- 1 large loss will not affect treaty performance and hence treaty renewal or reinstatement premium

(Max 3)

[13 Marks]

Q. 9)

a) Each of the 4 animal cohorts will have each of 3 age cohorts = total 12 cohorts

(1)

b) Highest frequency – Bird

Highest severity – Cat

(1)

c) Base frequency – New born Dog

Base severity – Above 5-year-old Cat

(1)

d) $\text{Frequency} = \text{Base Frequency} * \text{Frequency relativity of Animal} * \text{Frequency relativity of age}$ $\text{Severity} = \text{Base Severity} * \text{Severity relativity of Animal} * \text{Severity relativity of age}$

Animal	Age	Frequency	Severity	Burn Cost
Dog	0-2	0.10	57,000	5,700
Dog	2-5	0.05	52,250	2,351
Dog	Above 5	0.13	47,500	6,175
Cat	0-2	0.10	60,000	5,940
Cat	2-5	0.04	55,000	2,450

Cat	Above 5	0.13	50,000	6,435
Bird	0-2	0.25	58,800	14,700
Bird	2-5	0.11	53,900	6,064
Bird	Above 5	0.33	49,000	15,925
Rabbit	0-2	0.03	18,000	576
Rabbit	2-5	0.01	16,500	238
Rabbit	Above 5	0.04	15,000	624

(Max 12)

e) New born Bird is ~60 times risk compared to that of a 3-year-old Rabbit

(1)

f) Adjustments to burn cost:

- Expenses like operational costs, fixed expenses, future inflation, if any
- Sales expenses like commission and other sales costs
- Profit margin
- Smoothening for marketing
- Market testing of rates
- Loading for additional reinsurance costs

(Max 3)

[19 Marks]**Q. 10)****a)**

Claim No	Claim Amount	Sum Insured	% of SI = Claim Amt/SI
1234	60	600	10%
2345	70	175	40%
3456	150	750	20%
4567	30	120	25%
5678	100	200	50%
6789	80	100	80%
7900	120	160	75%
9011	100	500	20%
10122	80	100	80%
11233	50	250	20%

% of SI = X	% < X = A	X * A > X = B	C = A+B	D = Agg. C/max A
10%	60	78	138	16%
20%	360	96	456	54%
30%	390	135	525	63%
40%	460	152	612	73%
50%	560	140	700	83%
60%	560	168	728	87%
70%	560	196	756	90%
80%	840	-	840	100%
90%	840	-	840	100%
100%	840	-	840	100%

(9)

b) Burn cost for following structures:

- No deductible or limits = 840
- Deductible of 75 = 180
- Limit of 100 = 810
- Deductible of 50 & limit of 100 = 360

(Max 4)

c) No deductible or limits is best option for insured as it will have the highest level of risk transfer based on past claims data; whereas deductible of 50 & limit of 100 is best option for insurer as risk of the insurer is minimized

(Max 3)

[16 Marks]
